

ASSOCIATION BETWEEN FRIDGE OPENING FREQUENCY AND PROBABILITY OF FINDING SOMETHING NEW INSIDE

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ABSTRACT. Background: repeated refrigerator-door-opening behavior (“fridge-checking”) is common, but its relationship to the subjective probability of discovering something new inside remains uncharacterized. Methods: we conducted a 30-day prospective observational study of 42 households, logging daily fridge-opening frequency and self-reported discovery of novel contents. Results: discovery probability increased steadily with opening frequency, despite grocery deliveries being held constant across groups. Conclusion: these findings are consistent with an intermittent- reinforcement account of fridge-checking behavior, in which occasional genuine discoveries maintain a high rate of unrewarded checking.

Summary of changes

Comment	Type	Page	Section
R1-1	addition	2	Introduction
R1-2	replacement	3	Methods
R1-3	replacement	3	Methods
R1-5	replacement	2	Methods
R1-6	deletion	3	Methods
R1-7	addition	6	Results
R1-9	suppression	6	Results
R1-10	addition	2	Introduction
R1-10	addition	7	Conclusion
R2-1	addition	3	Methods
R2-2	addition	5	Results
R2-3	suppression	6	Results
R2-4	replacement	7	Discussion
R2-5	addition	4	Methods
R2-6	addition	6	Results
R2-7	deletion	5	Results
TUPPER-1	addition	3	Methods

1. INTRODUCTION

Refrigerators are widely believed to be static containers whose contents change only through the deliberate actions of their owners. This assumption has rarely

been tested directly. In this study, we investigate whether the frequency with which a refrigerator door is opened — a behavior we term fridge-checking — is associated with the subjective probability that its owner reports finding something new inside, independent of whether anything was actually added to the refrigerator during the observation period.

We define fridge-checking [, following the terminology introduced by \[1\]](#), as any refrigerator-door-opening event that is not immediately followed by either the removal or the addition of a food item. ^[R1-1]

We re-read this definition after the change above and confirm it still matches how fridge-checking events were actually coded during data collection. ^[IVANA-1]

[This comment from the first reviewer required a correction in two places in the manuscript — this sentence, in the introduction, and a matching one in the conclusion](#) — applies equally to both instances. ^[R1-10]

Anecdotally, household members report opening the refrigerator repeatedly over short periods despite reporting no expectation that its contents have changed. We refer to this informally as the “hope springs eternal” hypothesis, a fridge-specific corollary of general optimism bias. This preliminary report builds on our earlier observational work [2] by formally quantifying the relationship between opening frequency and discovery probability across a larger and more diverse sample of households.

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2. METHODS

2.1. Study population and design.

We recruited 42 households across three cities to participate in a 30-day prospective observational study. Eligible households were required to own at least one refrigerator, have at least one adult member willing to complete a daily fridge-opening log, and report no professional catering experience, to exclude expert fridge-checkers, who were assumed to behave differently.

Households were recruited via flyers posted in laundromats, grocery stores, and one particularly well-trafficked gym. ^[R1-4]

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^[R1-5]

	Low tertile	Medium tertile	High tertile
Household size, mean	1.8	2.4	2.6 3.1
Owens a pet, %	12	19	31
Fridge magnets, median	3	5	9
Snacks after 10pm, %	21	38	64

TABLE 1. Household characteristics by tertile of daily fridge-opening frequency.

As shown in Table 1, households in the high-opening tertile reported markedly more fridge magnets and a higher prevalence of after-hours snacking.

2.2. Statistical analysis (**updated**) ^[R2-1].

~~The association between opening frequency and discovery probability was assessed using ordinary logistic regression, with one observation per household. The association between opening frequency and discovery probability was assessed using generalized estimating equations with an exchangeable working correlation structure [3], to account for the repeated, within-household structure of the daily opening logs.~~ ^[R1-2]

The primary model

$$\cancel{P(\text{discovery}) = \beta_0 + \beta_1 \cdot \text{openings}} \quad (1)$$

$$\text{logit}(P(\text{discovery})) = \beta_0 + \beta_1 \cdot \text{openings} + u_{\text{household}} \quad (1)$$

is a naive linear probability model. ~~, where $u_{\text{household}}$ is a household-level random effect absorbing unmeasured, time-invariant fridge-checking tendencies.~~ ^[R1-3]

~~While implementing the change above, we also noted that the estimated household-level random effect variance, $\hat{\sigma}_u^2 = 0.42$, is worth reporting explicitly: it indicates non-trivial between-household heterogeneity in fridge-checking tendencies beyond what opening frequency alone explains.~~ ^[TUPPER-1]

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~~For completeness, calibrated discovery probabilities were also computed directly as~~

$$\cancel{\hat{p} = \text{expit}(\hat{\beta}_0 + \hat{\beta}_1 \cdot \text{openings})} \quad (2)$$

^[R1-6]

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aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

Theorem 2.2.1. *For any household refrigerator observed for a sufficiently long time, there exists at least one food item whose provenance cannot be established by any current member of the household.*

We propose to call this the Second Law of Fridge Thermodynamics. ^[R2-5]

Proof. Suppose, for contradiction, that every item in the refrigerator has a known provenance at every point in time. Then no item could ever transition from “recently purchased” to “mystery item” without an intervening act of forgetting. But forgetting is, empirically, unavoidable over a sufficiently long observation window [2]. This contradicts the assumption, and the claim follows. $\square$

3. RESULTS

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Figure 1 shows an apparent dose-response relationship between opening frequency and discovery probability, despite grocery deliveries being held constant across groups by design. ^[E1]

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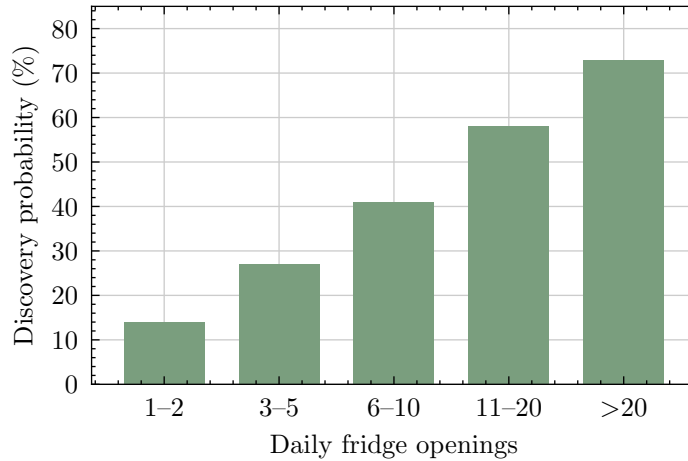


FIGURE 1. Reported probability of finding something new inside the refrigerator, by daily opening frequency (openings per day).

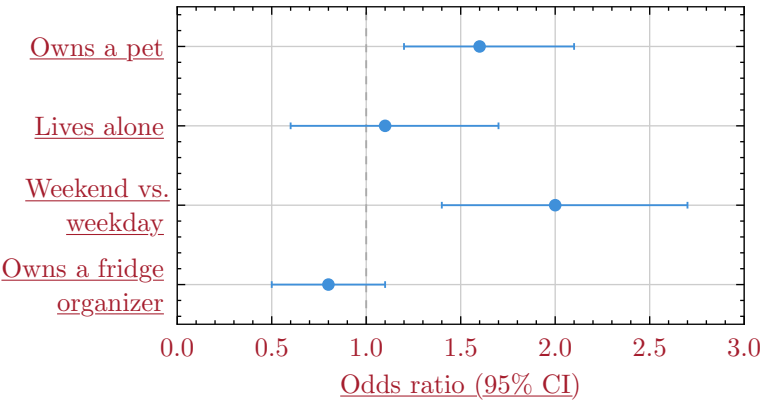


FIGURE 2. Subgroup odds ratios for the association between opening frequency and discovery probability.

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[R2-2]
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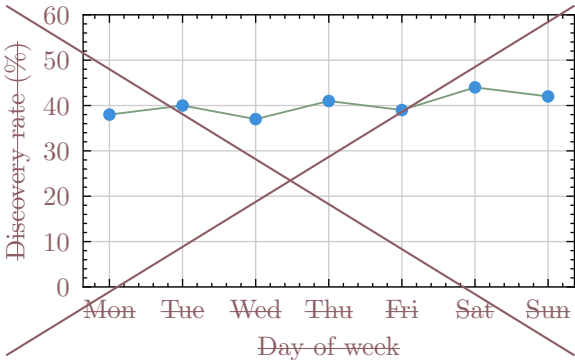


FIGURE 3. Discovery rate by day of week.

[R2-7]
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Category	Day 1	Day 30
Forgotten leftovers	61%	58%
Mystery Tupperware	24%	22%
Something green	9%	7%
Actually edible food	6%	13%
<u>Unidentified frozen object</u>	<u>=</u>	<u>3%</u>

TABLE 2. Fridge discovery categories, day 1 versus day 30 of observation.

transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At.

[Sub-analysis removed: the sock-drawer control group, including its summary figure and table]

[R2-3]

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[R2-6]

As shown in Table 2, the proportion of discoveries classified as “actually edible food” more than doubled over the observation period.

[Sensitivity analysis removed: results by handle-contamination threshold]

[R1-9]

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[A sensitivity analysis was conducted to assess robustness to unmeasured hunger confounding, following \[4\].](#) [R1-7]

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4. DISCUSSION

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~~These results suggest that fridge-checking behavior causes an increase in perceived discovery probability.~~ These results are consistent with an association between fridge-checking behavior and perceived discovery probability, though the observational design and the absence of a formal blinding procedure for refrigerator contents limit causal interpretation. ^[R2-4]

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5. CONCLUSION

Taken together, these findings suggest that the refrigerator door itself may function as an intermittent-reinforcement device, rewarding repeated checking with an occasional, genuinely novel discovery. ^[R1-8]

As already noted in the introduction, this conclusion applies equally to the paired instance of the same reviewer comment discussed there. ^[R1-10]

REFERENCES

1. Ware, T., Crumb, N.: Defining Fridge-Checking Behavior: A Delphi Consensus Study. *Annals of Domestic Science*. (2019)
2. Snackwell, I.: The Fridge-Checking Phenomenon: A Preliminary Report. *Journal of Kitchen Epidemiology*. (2020)
3. Estimator, G.E.: Generalized Estimating Equations for Clustered Snacking Data. *Journal of Applied Statistics*. (2017)
4. Brie, C.: Unmeasured Hunger Confounding in Observational Kitchen Studies. *Epidemiologic Methods*. (2022)

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